Cryptography Algorithms and Verilog Implementation 130330
Hard Coded Time Base Generator (Assignment 1)
Date: Sunday, March 8, 2026

Submission:	By not more than 2 people together
Deadline: Preparation work	There is no official preparation work for this lab. Please read through this whole paper and get familiar with the details and with what needs to be done...
Deadline: Final report	Sunday, March 22, 2026 (17:15) – via moodle
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment	

A. Time-Base Generator (Just a Reminder)

A **time-base generator** is a mechanism that creates a single pulse, **one clock cycle wide**, at regular pre-defined time periods (intervals). These generated pulses are used to enable counters or other synchronous mechanisms, periodically.

An example for the need of a time-base mechanism can be a time measurement system that has to count seconds, minutes and hours but, its only clock source comes from a 10MHz crystal. In order to advance the seconds counter, a pulse must be created from the 10MHz input clock, once every 10,000,000 cycles. The pulse width has to be only one cycle of the 10MHz clock.

Time-base generators can be either **hard coded** or **programmable.** This lab will exercise the hard coded time base generator.

B. Hard Coded Time-Base Generator

The hard coded time-base generator is a counter that generates a TC (Terminal Count) pulse after a fixed number of clock cycles. If the frequency of the input clock is changed, then the time-base generator is no longer accurate to how it was originally designed.

Block diagram of a Hard Coded Time-Base Generator is presented in the following diagram:

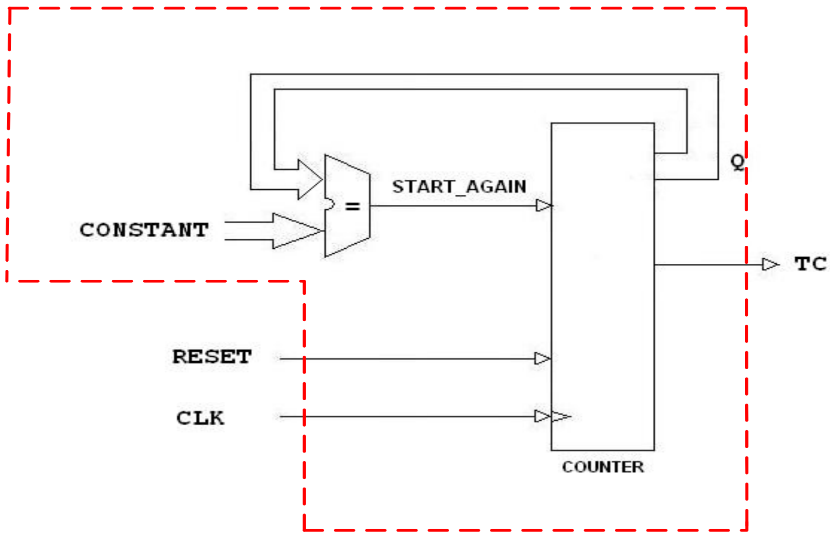

Figure 1: Hard Coded Time-Base Generator

The constant is internal (hard coded in the design source). The Q signal will be left as an internal signal. <u>The entity shall therefore contain only three signals: CLK, RESET and TC</u>.

Generally speaking, the hard coded time base generator is a counter. However, the counter's step (Q) is treated as an internal signal (i.e., it is usually not reflected as an output of that counter's entity). Clock, Reset and TC are the only I/Os usually required for that time base generator.

C. Design and Simulation Requirements for the Lab

1. **<u>Write the Verilog source code</u>** for a hard coded time-base generator based on your previous knowledge of writing counter's description:

   - Active high synchronous reset;
   - The TC (Terminal Count) output signal must be generated every 100 cycles of the input clock. (This can then act as a clock-enable input to another design/system/entity.)
   - Make sure that the "Q" signal is internal (and does not show in the entity)!

2. Prepare an appropriate **<u>Verilog testbench</u>** accordingly

D. In the Lab

- Simulate the design to show a number of TC pulses
- Synthesize the design to the Basys3's Xilinx-AMD Artix-7 device (use Precision), then Place and Route the design using Vivado (Post Synthesis Project). Use XDC file that is suitable for the Basys3 board.
- Synthesize and Place and Route the design to the Basys3's Xilinx-AMD Artix-7 device, use Vivado (RTL Project). Use XDC file that is suitable for the Basys3 board.

F. Your lab report must include the following:

1. The Verilog sources of your design and testbench;
2. Waveforms presenting the output pulse (TC) at the specified time (Verilog source level simulation).
3. Screenshots of "RTL Schematics", "Technology Schematics" of the two synthesis approaches (Precision and Vivado).
4. Screenshot of the "Device View" from Vivado, focused on the design.